

the first EtherCAT node integrated on an ARM Cortex-M controller. It enables the most compact designs, eliminating the need for a dedicated EtherCAT application specific integrated circuit (ASIC), external memory and crystal. Designer engineers benefit from reduced BOM cost and PCB space savings.

Freescal's S32K MCU also supports CAN FD apart from the new FlexIO-configurable peripheral that enables implementation of future communication protocols. It also allows expanding channels to existing on-chip hardware protocol controllers. According to their press release, you can now "Implement future communication protocols that are not yet invented."

Targeting the rapid growth of the IoT market, Atmel is combining its ultra-low-power SAM G series of MCUs designed for wearable and sensor-hub management, and their secure ultra-low-power SmartConnect WILC1000 Wi-Fi solution.

STMicroelectronics's new SPC58NE is one among the first chips targeting the automotive space, to come out on STMicroelectronics' 40nm process node. It combines multiple dual-issue cores with up to 6MB flash and 768kB internal RAM memory, eight CAN interfaces and an optimised peripheral set based on the end application. The multiple cores ensure redundancy in these applications to meet the safety and security demands of vehicle manufacturers.

Better peripheral and sensor integration

"More peripherals are being supported on the sensor side to monitor and control, while sensor fusion is now included to help in computing and amalgamating data from various sensors as well as for encryption. Some of these are LCD display controllers, HDMI interfaces, camera interface Mobile Industry Processor Interface and touch drivers," says Kamat.

LAPIS Semiconductor, a ROHM Group company, has recently announced the ML620530/540/550 fam-

ily of 16-bit low-power MCUs that integrate 256kB flash memory along with a proprietary (these are not ARM cores but LAPIS' original RISC CPU U16) 16-bit low-power CPU core with a newly-developed successive approximation AD converter featuring 1.25µs conversion speed and high accuracy ($\pm 1\%$) oscillation circuit.

These are said to help in efficient processing of multiple analogue inputs (that is, from sensors) and operations requiring high oscillation accuracy such as mutual data communication between devices without the need for an external oscillator, making it possible to configure simpler systems. Specifically, this allows support for applications that utilise networks in the future, such as serial communication that enables wireless connection with HEMS/BEMS equipment and multichannel analogue input for motion, temperature, pulse and other biological sensors used in healthcare equipment, as well as conventional sensing and actuation that contribute to greater energy savings and convenience in today's home appliances and electronics.

Atmel SMART SAM G54/G55 comes with more SRAM and integrated sensor-fusion algorithms to make it easier to make sense of sensor data.

Atmel SMART SAM DA1 ARM Cortex-M0+ embeds a peripheral touch controller that offers high performance and low-power capacitive-touch support in compact QFN 5mm x 5mm package with wettable flanks for automated optical inspection.

With capacitive now becoming almost a standard human-machine interface, the RX231 group from Renesas provides capacitive-touch-sensor capabilities equipped with Renesas' second-generation Ca Touch technology. With support for up to 24 touch-keys for self-capacitance mode and up to 144 keys for mutual-capacitance mode, RX231 MCUs can support LEDs, buzzers and communication functions in more basic touch systems such as blood pressure monitors (typically nine keys) to systems

with more elaborate requirements, like security keypads (19 keys) and ovens (25 keys). This functionality combines high sensitivity with a high level of noise tolerance, allowing use even when the panel is wet or when the operator is wearing gloves.

MCUs versus SoCs

Growing support for more peripherals, larger memories and smarter CPU features among MCU families show just how much the MCU has changed over the course of a decade. As the amount of integration increases, we see a blur in the line separating the MCUs from systems on chips (SoCs). "Moreover, technologies that were previously seen in microprocessors, like interfaces and power, are now increasingly being seen in MCUs," explains Sathesh Kumar Murugan, project manager, Embitel Technologies.

"Security has become a key vector that gets integrated, with the Internet and cloud being a part of almost every device. Chips have had flash as integrated memory, but since the last year, we have had MCUs with embedded FRAMs. It lets us offer devices that can run on very low power. These are extremely fast, too. The integrated FRAM on the MCU eliminates the need for external EEPROM, while offering faster writes and flexibility in code and data allocation, thus reducing board space and cost, while enhancing performance and ease of programming," adds Ganapathy.

Alexandru Voica, senior technology marketing specialist, Imagination Technologies, says, "A lot of the features previously available only for high-end CPUs are slowly but surely trickling down into the MCU space. Recently launched MCUs are getting DSP/SIMD functionality, virtualisation support and advanced security features. For example, our M-class M5150 MCU integrates all of the above, plus an MMU and cache controllers. This enables it to run a full-blown OS like Linux. M5150 is probably the most feature-rich MCU-class CPU IP available today. ●